

Guidelines:

You should work on these exercises from the third to the sixth week of the course. At the end of the sixth week, you will take a "Homework Oriented Exam" (see Canvas/Datanose for details). The exam will include exercises similar to those in this homework set, possibly with minor variations or additional questions.

We will not provide a solution sheet for this homework set. However, you are encouraged to discuss any questions you have about the exercises with your TAs during the Friday seminars.

When working on your answers, please remember the following points:

- Your solutions should include the intermediate steps of your reasoning and derivations. Answers without justification will not receive full marks in the exam. Also, remember to simplify your final answers as much as possible.
- We use the numerator layout convention for multivariate derivatives (i.e., the derivative of a function $f : \mathbb{R}^m \rightarrow \mathbb{R}^n$ has shape $n \times m$).
- When computing multivariate derivatives, you should first compute the derivative for arbitrary components (e.g., $\frac{\partial f_i}{\partial x_j}$). After deriving this general form, assemble the final answer as a matrix/vector and simplify. This is what we refer to as *index notation*.

Hand-in Exercises

You may submit the following exercises on the dates below to receive feedback from your TAs:

- Friday, September 25th: Exercise 2
- Friday, October 2nd: Exercise 6

Please note:

- Only handwritten, paper submissions will be accepted.
- Please write your name and group (1,2,3/4,5,6/...) in the top right corner.
- Submission is voluntary and will not affect your grade. Please ensure your submitted solutions are entirely your own work, as this allows us to provide the most helpful feedback.

Exercise 1: Naive Bayes Modeling of Personality Types

Naive Bayes (NB) is a particular form of classification that makes strong independence assumptions regarding the features of the data, conditional on the classes. Specifically, NB assumes each feature is independent given the class label. In contrast, probabilistic generative models with full-covariance Gaussians incorporate correlations between all input features.

If correlated features are treated independently, the evidence for a class will be overcounted. However, Naive Bayes is very simple to construct, because by ignoring correlations the class-conditional likelihood is a product of D univariate distributions:

$$p(\mathbf{x} \mid \mathcal{C}_k) = \prod_{d=1}^D p(x_d \mid \mathcal{C}_k).$$

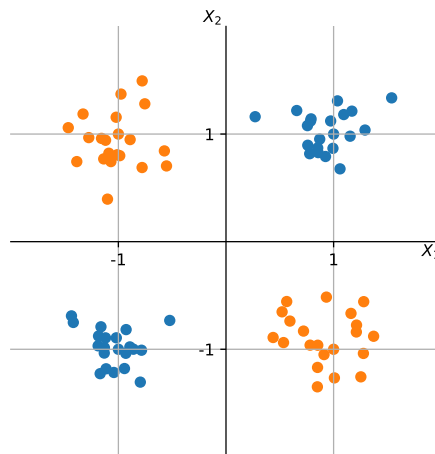
Consider classifying people into K personality types $\mathcal{C}_k$, where $\mathcal{C}_K$ means “ambiguous”. For each of N people, we observe a vector $\mathbf{x}_n \in \mathbb{R}_{\geq 0}^D$, where x_{nd} is the amount of money person n spends per month in life area d . The target matrix $\mathbf{T}$ has rows $\mathbf{t}_n^T = (t_{n1}, \dots, t_{nK})$, one-hot encoded. Assume $p(\mathcal{C}_k) = \pi_k$ and $\sum_{k=1}^K \pi_k = 1$. We model spending in life area d conditioned on class $\mathcal{C}_k$ with an exponential distribution with parameter λ_{dk} :

$$p(\mathbf{x} \mid \mathcal{C}_k, \lambda_{1k}, \dots, \lambda_{Dk}) = \prod_{d=1}^D \lambda_{dk} e^{-\lambda_{dk} x_d}.$$

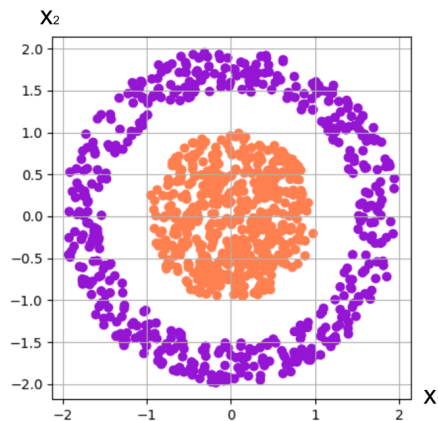
- Write down the data likelihood $p(\mathbf{T}, \mathbf{X} \mid \mathbf{\Lambda})$ without the Naive Bayes conditional independence assumption at first. Then derive the likelihood for the general K -class Naive Bayes classifier, stating where you use the product rule and the Naive Bayes assumption. Write the likelihood in terms of $p(x_{nd} \mid \mathcal{C}_k)$, without assuming the exponential distribution.
- How does the number of parameters change if you make use of the Naive Bayes assumption? Why is the assumption called naive? Can you think of an example in which this assumption does not hold?
- Write down the data log-likelihood $\ln p(\mathbf{T}, \mathbf{X} \mid \mathbf{\Lambda})$ for the given exponential likelihood model.
- Solve for the MLE estimator of λ_{dk} . Express in your own words how the result can be interpreted.
- Specify $p(\mathcal{C}_1 \mid \mathbf{x})$ for the general K -class Naive Bayes classifier in terms of the prior and class-conditional distributions. Specify your final answer in terms of π and λ .
- Write $p(\mathcal{C}_1 \mid \mathbf{x})$ for the exponential model explicitly in terms of the modeling choices.
- Write down as an inequality the region for which $\mathbf{x}$ is predicted to be in class $\mathcal{C}_1$ compared to all other classes in terms of the posterior from the previous question. Subsequently, specify the inequality as a linear inequality of the form $\mathbf{x}^T \mathbf{a} > c$.
- Is the region where $\mathbf{x}$ is predicted to be in $\mathcal{C}_1$ convex? Why?

Exercise 2: Binary Classification

Suppose you are given the following dataset:



- (a) For each of the following methods, explain why they can or cannot classify the dataset.
- (a) Logistic regression with linear features, $y = \sigma(\mathbf{w}^T \mathbf{x})$.
 - (b) Logistic regression with nonlinear basis functions, $y = \sigma(\mathbf{w}^T \phi)$.
 - (c) Multilayer perceptron with one hidden layer.
- (b) Can the dataset above be classified using Naive Bayes? Remember that we are free to set the marginal densities $p(x_i | C_j)$ as we please. If yes, provide the marginal densities. If no, prove it. *Hint: Consider the subset $\{(-1, 1), (1, 1), (1, -1), (-1, -1)\}$.*
- (c) Now consider the following dataset. Can it be classified using Naive Bayes? How is your reasoning different from the one in the previous question? *Hint: Can you think of a distribution that fits this data? You may assume x_1 and x_2 to be continuous.*



- (d) What is the main difference between logistic regression with nonlinear basis functions ϕ and multilayer perceptrons in terms of ϕ ?

Exercise 3: Regularized Logistic Regression

Consider logistic regression for K classes with N training vectors $\{\mathbf{x}_n\}_{n=1}^N$, mapped to feature vectors

$$\phi_n = \phi(\mathbf{x}_n) = (\phi_0(\mathbf{x}_n), \phi_1(\mathbf{x}_n), \dots, \phi_{M-1}(\mathbf{x}_n))^T,$$

where $\phi_0(\mathbf{x}_n) = 1$. Each input has target vector $\mathbf{t}_n = (t_{n1}, \dots, t_{nK})^T$, with $t_{nk} = 1$ if $\mathbf{x}_n \in \mathcal{C}_k$ and 0 otherwise. Collect the targets in $\mathbf{T}$ and features in Φ . The posterior class probabilities are modeled by

$$p(\mathcal{C}_k | \phi, \mathbf{w}_1, \dots, \mathbf{w}_K) = y_k(\phi) = \frac{\exp(a_k)}{\sum_{j=1}^K \exp(a_j)},$$

where $a_k = \mathbf{w}_k^T \phi$. Assume the Gaussian prior

$$p(\mathbf{w}_1, \dots, \mathbf{w}_K | \alpha) = \prod_{k=1}^K \mathcal{N}(\mathbf{w}_k | \mathbf{0}, \alpha^{-1} \mathbf{I}).$$

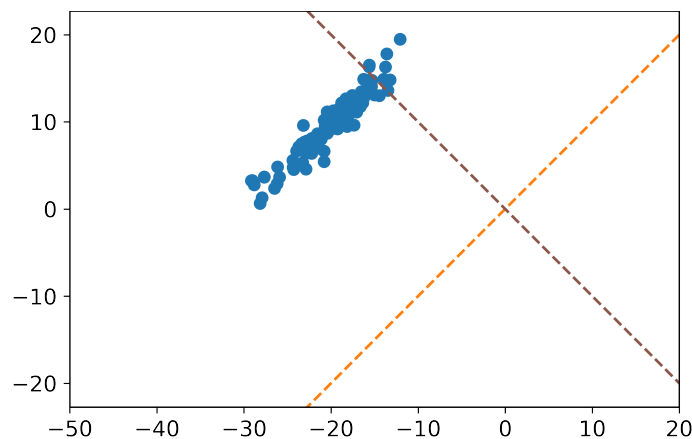
- (a) Write down the likelihood $p(\mathbf{T} | \Phi, \mathbf{w}_1, \dots, \mathbf{w}_K)$ as a product over N and K , and then write down the log-likelihood.
- (b) Write down the explicit form of the prior and its logarithm. How does computing the logarithm help during computations?

- (c) Write down the posterior $p(\mathbf{w}_1, \dots, \mathbf{w}_K \mid \Phi, \mathbf{T}, \alpha)$ by applying Bayes' rule.
- (d) Show that obtaining a MAP estimate for $\mathbf{w}_1, \dots, \mathbf{w}_K$ is equivalent to regularized logistic regression for K classes, minimizing

$$-\sum_{n=1}^N \sum_{k=1}^K t_{nk} \log y_k(\phi_n) + \frac{\alpha}{2} \sum_{k=1}^K \sum_{m=0}^{M-1} |w_{km}|^2.$$

- (e) Suppose you are confident that the weights should lie around 0. What variable changes in the equation from the previous part, and why? Why does this make the weights closer to 0?
- (f) Now assume you are very unsure of the values the weights should take. Prove that the more uncertain you are, the closer the MAP solution approaches the MLE solution.
- (g) Derive the gradient of the log-likelihood $\nabla_{\mathbf{w}_j} \log p(\mathbf{T} \mid \Phi, \mathbf{w}_1, \dots, \mathbf{w}_K)$. Your final answer should be a single sum over n . You may use $\frac{\partial y_k}{\partial a_j} = y_k(\mathbb{I}_{kj} - y_j)$.
- (h) Derive the gradient of the log-prior and the gradient of the log-posterior with respect to $\mathbf{w}_j$.

Exercise 4: Principal Component Analysis



There are many ways of defining PCA. Here we will focus on three of them. Remember that we are given $\mathbf{x}_1, \dots, \mathbf{x}_n \in \mathbb{R}^d$ and we want to find an orthogonal projection matrix $P \in \mathbb{R}^{d \times d}$ of rank k . The goal of PCA is to get projections $P\mathbf{x}_1, \dots, P\mathbf{x}_n \in \mathbb{R}^d$ that “retain most of the original information”. This is hand-wavy, so let us formalize it in three different ways.

- (A) Maximize the scatter:

$$\max_P \sum_{i=1}^n \|P\mathbf{x}_i - P\bar{\mathbf{x}}\|^2.$$

- (B) **After centering**, minimize the reconstruction error:

$$\min_P \sum_{i=1}^n \|(\mathbf{x}_i - \bar{\mathbf{x}}) - P(\mathbf{x}_i - \bar{\mathbf{x}})\|^2.$$

- (C) Preserve pairwise distances as much as possible. A projection matrix is a contraction, so $\|P\mathbf{y}\|^2 \leq \|\mathbf{y}\|^2$. Thus, all pairwise distances $\|\mathbf{x}_i - \mathbf{x}_j\|^2$ become smaller after projection, and we want to make this gap as small as possible:

$$\min_P \sum_{i,j=1}^n \|\mathbf{x}_i - \mathbf{x}_j\|^2 - \|P\mathbf{x}_i - P\mathbf{x}_j\|^2.$$

You will prove that the three formulations are equivalent.

- (a) For formulation (A), prove that it is equivalent to

$$\max_P \sum_{i=1}^n (\mathbf{x}_i - \bar{\mathbf{x}})^T P (\mathbf{x}_i - \bar{\mathbf{x}}).$$

- (b) For formulation (A), prove that it is equivalent to

$$\max_P \text{Tr}(S_1 P).$$

What is S_1 in this case?

- (c) For formulation (B), why do we want to center the data before minimizing the reconstruction error? Use the scatter plot above to make your point.
(d) For formulation (B), prove that it is equivalent to

$$\min_P \sum_{i=1}^n (\mathbf{x}_i - \bar{\mathbf{x}})^T (I - P) (\mathbf{x}_i - \bar{\mathbf{x}}).$$

- (e) For formulation (B), prove that it is equivalent to

$$\min_P \text{Tr}(S_2 (I - P)).$$

What is S_2 in this case?

- (f) For formulation (C), prove the contraction property of projection matrices. *Bonus*.
(g) For formulation (C), prove that it is equivalent to

$$\min_P \sum_{i,j=1}^n (\mathbf{x}_i - \mathbf{x}_j)^T (I - P) (\mathbf{x}_i - \mathbf{x}_j).$$

- (h) For formulation (C), prove that it is equivalent to

$$\min_P 2n \text{Tr}(S_3 (I - P)).$$

What is S_3 in this case?

- (i) Finally, show that the three formulations are equivalent.
(j) The solution relies on the eigendecomposition of the covariance matrix $S = V D V^T$.
- What is the relationship between V and the dotted lines on the scatter plot above?
 - Can you express P in terms of V ? You can just state the final solution.

- Plot the projected points on the scatter plot above, assuming $k = 1$.

Exercise 5: Probabilistic PCA – A General Latent Space Distribution

Principal Component Analysis (PCA) is an often used technique for dimensionality reduction. In this approach, we linearly project data onto a lower-dimensional subspace. This approach can be extended to a latent variable model named probabilistic PCA. Probabilistic PCA has several advantages over conventional PCA:

- We can associate a likelihood function with probabilistic PCA, allowing direct comparison with other probabilistic density models.
- Probabilistic PCA can model class-conditional densities and can therefore be used in classification problems.
- We can run the model generatively to provide samples from the modeled distribution.

Probabilistic PCA is an example of the linear-Gaussian framework, where both marginal and conditional distributions are Gaussian. Define a latent variable $\mathbf{z}$ corresponding to the principal-component subspace, with conditional distribution

$$p(\mathbf{x} | \mathbf{z}) = \mathcal{N}(\mathbf{x} | \mathbf{W}\mathbf{z} + \boldsymbol{\mu}, \sigma^2 \mathbf{I}).$$

The prior distribution over $\mathbf{z}$ is usually

$$p(\mathbf{z}) = \mathcal{N}(\mathbf{z} | \mathbf{0}, \mathbf{I}).$$

In this case, the marginal distribution is

$$p(\mathbf{x}) = \mathcal{N}(\mathbf{x} | \boldsymbol{\mu}, \mathbf{W}\mathbf{W}^T + \sigma^2 \mathbf{I}).$$

Suppose instead that the latent prior is a general Gaussian,

$$p(\mathbf{z}) = \mathcal{N}(\mathbf{z} | \mathbf{m}, \boldsymbol{\Sigma}).$$

We will show that, by redefining model parameters, this prior gives an identical family of marginal distributions $p(\mathbf{x})$ for any valid $\mathbf{m}$ and $\boldsymbol{\Sigma}$.

- Express $\mathbf{z}$ as $\mathbf{z} = \mathbf{m} + \boldsymbol{\epsilon}_z$, with $\boldsymbol{\epsilon}_z \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma})$. Write a similar expression for $\mathbf{x} = \mathbf{W}\mathbf{z} + \boldsymbol{\mu} + \boldsymbol{\epsilon}_x$, and explain what distribution $\mathbf{x}$ follows. From which distribution is $\boldsymbol{\epsilon}_x$ sampled?
- Find the expected value of $\mathbf{x}$ using linearity of expectation.
- Find the covariance matrix of $\mathbf{x}$.
Hint: $\text{Var}[\mathbf{A}\mathbf{Y}] = \mathbf{A} \text{Var}[\mathbf{Y}] \mathbf{A}^T$.
- To show that the general Gaussian prior still leads to an identical model, write $p(\mathbf{x})$ in the form

$$p(\mathbf{x}) = \mathcal{N}(\mathbf{x} | \tilde{\boldsymbol{\mu}}, \tilde{\mathbf{W}}\tilde{\mathbf{W}}^T + \sigma^2 \mathbf{I}).$$

Find suitable expressions for $\tilde{\boldsymbol{\mu}}$ and $\tilde{\mathbf{W}}$.

Exercise 6: Mixture of Experts

In class, you discussed mixture models as a way to perform unsupervised learning tasks, *e.g.*, clustering. Mixture models can similarly be used for supervised learning tasks. In this question, we discuss mixtures of experts (MoEs), a model that softly partitions the input space and learns a supervised model for each area.

Consider K expert models for a dataset of N data points $\{(\mathbf{x}_1, y_1), \dots, (\mathbf{x}_N, y_N)\}$, where $\mathbf{x}_n \in \mathbb{R}^D$ and y_n is the ground-truth label for $\mathbf{x}_n$. Let z_n denote a categorical random variable indicating which expert model is active, and let $\mathbf{z}_n$ be the one-hot encoding of z_n . Let $\Theta \in \mathbb{R}^{D \times K}$ contain the expert parameter vectors as columns. Assume each y_n is continuous on $[0, \infty)$ and follows an exponential distribution. Expert k has the predictive model

$$p(y_n | \mathbf{x}_n, \mathbf{z}_n, \Theta) = p(y_n | \mathbf{x}_n, z_n = k, \theta_k) = \text{Exponential}(y_n | \lambda = \exp(\theta_k^T \mathbf{x}_n)),$$

where

$$\text{Exponential}(y | \lambda) = \lambda \exp(-\lambda y), \quad y \geq 0.$$

The routing mechanism determines how relevant each expert is for a specific datapoint:

$$p(z_n = k | \mathbf{x}_n, \Phi) = \pi_{nk} = \frac{\exp(\phi_k^T \mathbf{x}_n)}{\sum_{j=1}^K \exp(\phi_j^T \mathbf{x}_n)},$$

where $\Phi = [\phi_1, \dots, \phi_K] \in \mathbb{R}^{D \times K}$.

- The routing output falls between 0 and 1. Write down the formula to decide each element z_{nk} of $\mathbf{z}_n$, assuming you want the most relevant expert to be active for datapoint $\mathbf{x}_n$.
- Since we have no prior information about which expert is responsible for a particular prediction, marginalize over all possible experts. Write down the likelihood $p(\mathbf{y} | \mathbf{X}, \Theta, \Phi)$ and the log-likelihood of the entire dataset.
- Write down the posterior probability r_{ni} of expert i producing label y_n for datapoint n , also called the responsibility of expert i for datapoint n .
- Take the derivative of the log-likelihood with respect to the parameters of each expert θ_i and the routing parameters ϕ_i . Do not substitute explicit distributions; express your answer in terms of $p(y_n | \mathbf{x}_n, z_n, \theta_i)$, $p(z_n = k | \mathbf{x}_n, \Phi)$, and responsibilities r_{ni} .
- Insert the explicit expressions for the probability distributions and compute the final derivatives for θ_i and ϕ_i .
- Write down an iterative algorithm that maximizes the log-probability of the data by jointly optimizing Θ and Φ . Use appropriate convergence criteria.
- Suppose that instead of having the prediction for $\mathbf{x}_n$ determined by one expert, we mix the predictions of all experts. To compute the final prediction $\hat{y}_n$, weigh the prediction of each expert by its relevance. Write down the formula, denoting the prediction of expert k for datapoint n by $\hat{y}_{nk}$.